PHY202: Physics II: Exam 6
Dr. Deskins

Excellent!

$\frac{12}{12} \Rightarrow \underline{100}\%$

Name:_

These questions assess your understanding of the material from Chapters 25, 26 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Calculate the power of the eye when viewing an object 24. km away. Take the lens-to-retina distance to be 2.5 cm.

$P = \frac{1}{do} + \frac{1}{di}$

ANSWER: 40.0D

$P = \frac{1}{24 \times 10^3} + \frac{1}{0.025} = 40.0\,D$

2. Calculate the speed at which light travels in a particular material given that the index of refraction of the material is 1.3.

$n = \frac{c}{v} \quad v = \frac{c}{n} = \frac{3 \times 10^8}{1.3} = 2.31 \times 10^8 \, m/s$

ANSWER: $22.31 \times 10^8 m/s$

3. Suppose that you take a magnifying glass out on a sunny day and you find that it concentrates sunlight to a small spot 8.1 cm away from the lens. What is the power of the lens?

ANSWER: 12.3D

$P = \frac{1}{f} \quad P = \frac{1}{0.081} = 12.3D$

4. What is the size of the image on the retina of a 0.77 mm-diameter strand of hair that is held at arm's length (40. cm) away? Take the lens-to-retina distance to be 1.5 cm.

ho, do, di

ANSWER: $-2.89 \times 10^{-5} m$

$\frac{hi}{ho} = -\frac{di}{do}$

$di = 0.015$
$ho = 0.77 \times 10^{-3}$
$do = 0.4$

$hi = -\frac{di\,ho}{do}$

$hi = -\frac{(0.015)(0.77 + 1c^{-3})}{0.4} = -2.89 \times 10^{-5} m$

+4

5. What power of spectacle lens is needed to correct the vision of a near-sighted person whose far point is 38. cm? Assume the spectacle lens is held 1.5 cm away from the eye by eyeglass frames.

$d_i = -(0.38 - 0.015) = -0.365$

ANSWER: -2.74 D

$d_o = \infty$

$P = \cancel{\frac{1}{d_o}}^{0} + \frac{1}{-0.365} = -2.74$

6. You shine a laser through air at an incident angle of 32.° into an unknown material and measure the angle of refraction to be 11.°. What is the unknown material's index of refraction?

$n_1 \sin \theta_1 = n_2 \sin \theta_2$

ANSWER: 2.78

$1 \sin 32 = n_2 \sin 11$

$n_2 = \frac{\sin 32}{\sin 11} = 2.78$

7. Calculate the power of the eye when viewing an object 44. cm away. Take the lens-to-retina distance to be 2.6 cm.

ANSWER: 40.73 D

$P = \frac{1}{0.44} + \frac{1}{0.026} = 40.73$

8. Suppose that you would like to find the angle of refraction in a material. You shine a laser through air at an incident angle of 29.° into the material which has an index of refraction of 1.4. Calculate the angle of refraction in the material.

$1 \sin 29 = 1.4 \sin \theta$

ANSWER: 20.26°

$\frac{\sin 29}{1.4} = \sin \theta \quad \theta = \sin^{-1}\left(\frac{\sin 29}{1.4}\right) = 20.26$

+4

d_o 0.39 f 0.41

9. A clear-glass light bulb is placed 0.39 m from a thin convex lens that has a focal length of 0.41 m. Calculate the distance from the lens to the location of the produced image.

$\frac{1}{d_i} = \frac{1}{f} - \frac{1}{d_o}$

$\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$

ANSWER: $-7.995m$

$d_i = \left(\frac{1}{f} - \frac{1}{d_o}\right)^{-1}$

$\left(\frac{1}{0.41} - \frac{1}{0.39}\right)^{-1} = -7.995m$

10. Calculate the critical angle for light traveling in a material that is surrounded by air, given that the material has an index of refraction of 1.4.

$1 \sin 90 = 1.4 \sin \theta$

ANSWER: $45.58°$

$\frac{1}{1.4} = \sin\theta$

$\theta = \sin^{-1}\left(\frac{1}{1.4}\right) \approx 45.58$

11. A scientist at a natural-history museum is holding a specimen 8.0cm from a thin convex magnifying glass of focal length 19.cm. What magnification is produced?

$m = \frac{-d_i}{d_o}$ $d_i = \left(\frac{1}{0.19} - \frac{1}{0.08}\right)^{-1} = -0.138$

ANSWER: 1.73

$\frac{-(-0.138)}{0.08} = 1.73$

12. A clear-glass light bulb is placed 0.30 m from a thin convex lens that has a focal length of 0.12 m. Calculate the magnification of the produced image.

$m = \frac{-d_i}{d_o}$ $d_i = \left(\frac{1}{0.12} - \frac{1}{0.3}\right)^{-1} = 0.2$

ANSWER: -0.67

$m = \frac{-0.2}{0.3} = -0.67$

+4

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1. $P = 40.0\,\mathrm{D} \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f},\ P = \frac{1}{f}$
2. $v = 2.31 \times 10^8\,\frac{\mathrm{m}}{\mathrm{s}} \because n = \frac{c}{v}$
3. $P = 12.3\,\mathrm{D} \because P = \frac{1}{f}$
4. $h_i = -2.89 \times 10^{-5}\,\mathrm{m} \because \frac{h_i}{h_0} = -\frac{d_i}{d_0} = m$
5. $P = -2.74\,\mathrm{D} \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f},\ P = \frac{1}{f}$
6. $n_2 = 2.78 \because n_1 \sin\theta_1 = n_2 \sin\theta_2$
7. $P = 40.7\,\mathrm{D} \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f},\ P = \frac{1}{f}$
8. $\theta_2 = 20.3^\circ \because n_1 \sin\theta_1 = n_2 \sin\theta_2$
9. $d_i = -8.00\,\mathrm{m} \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f}$
10. $\theta_c = 45.6^\circ \because n_1 \sin\theta_1 = n_2 \sin\theta_2$
11. $m = 1.73 \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f},\ \frac{h_i}{h_0} = -\frac{d_i}{d_0} = m$
12. $m = -0.667 \because \frac{1}{d_0} + \frac{1}{d_i} = \frac{1}{f},\ \frac{h_i}{h_0} = -\frac{d_i}{d_0} = m$